NCP shadow reachability: exact minimum word length

THM-COMMUTATOR-KERNEL1 Lemma 2(b) / TASK-20260822-3dd086 / Lean 4.35.0-rc1

Reachable at degree d iff L1(a) + m(e,a) <= d

L1(a) = sum |a_i|; T+ lists positive counts; T- lists negative counts; v is negation.

e = +1, all a_i nonnegative

m = 0

word: T+

e = -1

m = 1

word: T- v T+

e = +1, some a_i negative

m = 2

word: T+ v T- v

Lower bound: L1(a) <= number of translations; parity and any negative count force m(e,a) negations.

The construction reaches equality for every integer vector a, every factor-base size B, and both allowed signs.